Stage 4 Report: Diffusion and Project Progress

Image Super-Resolution with Deep Generative Models · Computer Vision 2026 · 27 Oct 2026 Team: Iuliia Kurnaeva, Alena Petrenko, Amaliya Kharisova, Dmitrii Kuznetsov

Repository: https://github.com/Ephy01/Image-Super-Resolution-with-Deep-Generative-Models

1. Completed experiments

We have evaluated four approaches: bicubic interpolation, Real-ESRNet, Real-ESRGAN and, this week, **ResShift v3**. No model is trained or fine-tuned. The current comparison uses the same 64 dev images (20 objects, 20 faces and 24 text images), downsampled from 512×512 to 128×128 and compressed at JPEG quality none, 50, 30 and 10: **256 inputs per method**.

- **Diffusion experiment.** ResShift uses the official pretrained model and autoencoder, a pinned implementation, four sampling steps and ×4 upscaling. It ran in Google Colab on a Tesla T4 with float32 weights and float16 autocast, batch size 1. Each image has a fixed sampling seed, reused across JPEG levels; the global and bootstrap seed is 42.
- **Reproducibility.** The executed notebook loads an archive of the original prepared images. Only ResShift is run; the earlier three-method baseline tables are reused. Input checksums, model versions, weight checksums, environment and timing are saved. The output PNGs reproduce the recorded PSNR and SSIM values.

2. Preliminary comparison (dev split, 64 images)

PSNR and SSIM on the Y channel with a 4-pixel border crop; LPIPS with AlexNet. Mean over images; best mean per content type and JPEG condition in bold.

Content	Method	PSNR ↑, no JPEG	SSIM ↑	LPIPS ↓	PSNR ↑, JPEG 10	SSIM ↑	LPIPS ↓
Objects	Bicubic	25.62	0.689	0.488	22.77	0.543	0.767
Objects	Real-ESRNet	**26.31**	**0.737**	0.376	**22.97**	**0.582**	0.613
Objects	Real-ESRGAN	24.44	0.682	0.255	21.80	0.547	**0.451**
Objects	ResShift	25.47	0.701	**0.212**	22.26	0.528	0.456
Faces	Bicubic	33.40	0.879	0.289	28.25	0.768	0.647
Faces	Real-ESRNet	**33.76**	**0.894**	0.238	**28.72**	**0.802**	0.456
Faces	Real-ESRGAN	31.05	0.859	0.159	27.65	0.776	0.342
Faces	ResShift	32.26	0.862	**0.108**	28.01	0.761	**0.322**
Text	Bicubic	16.46	0.707	0.436	15.12	0.581	0.639
Text	Real-ESRNet	19.31	0.838	0.186	**15.67**	**0.690**	0.408
Text	Real-ESRGAN	17.83	0.818	0.107	14.38	0.680	0.212
Text	ResShift	**19.76**	**0.858**	**0.085**	14.96	0.669	**0.195**

Without JPEG, ResShift improves on Real-ESRGAN by 1.03 dB PSNR on objects, 1.21 dB on faces and 1.92 dB on text, while reducing LPIPS in all three categories. Real-ESRNet retains the highest PSNR and SSIM on photographs. ResShift averages **0.909 s per image** on the T4 after three warm-up runs; earlier baseline timings were measured on another GPU and are not directly comparable.

3. Observations and difficulties

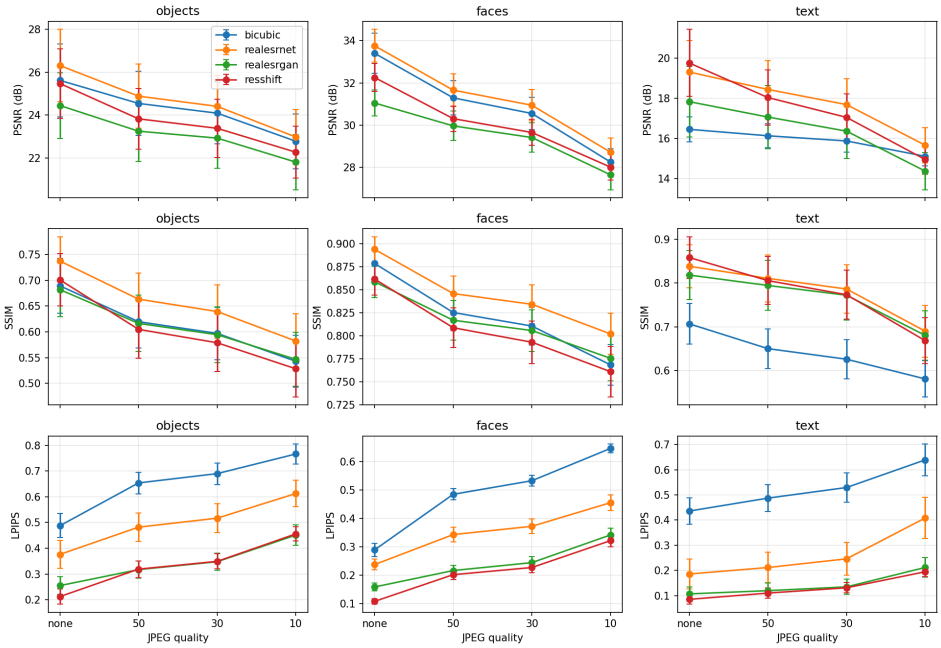
- **The advantage depends on degradation.** At JPEG 10, ResShift still has higher mean PSNR than Real-ESRGAN, but lower SSIM in every category. On objects, its LPIPS is slightly worse (0.456 vs. 0.451).
- **Perceptual scores do not establish content accuracy.** The JPEG 10 examples contain altered facial details and sharp, incorrect text strokes. Lower LPIPS does not demonstrate correct letters or preserved face identity.
- **Identical inputs matter.** An earlier Colab run regenerated the data and ran all methods. It is separate from this comparison, which uses the archived original inputs and baseline values. The earlier test results are retained as exploratory; the test split has already been accessed.

4. Remaining work

- Add MUSIQ, OCR character/word error rates by text size, and ArcFace similarity for faces.
- Repeat ResShift with several sampling seeds to measure output and metric variability.
- Benchmark all learned models on the same GPU and record peak memory use.
- Review confidence-interval reporting, freeze the evaluation settings, and complete the final comparison on the original 276-image test split, documenting the earlier exploratory test access.
- Consolidate the final quantitative tables, visual examples and project report.

5. Figures

Metrics against input JPEG quality. Error bars retain the project's existing 95% bootstrap half-width convention (10,000 resamples); interval reporting will be reviewed before the final evaluation.



ResShift at JPEG quality 10: low-resolution input, reconstructed image and reference. These are full images; sharper text strokes do not imply that the original characters were recovered.

